

E-commerce Sales and Profitability Analysis

ShopSphere – Data Analytics Project Documentation

Project Title: E-commerce Sales and Profitability Analysis
Company: ShopSphere
Domain: Retail / E-commerce Analytics
Prepared For: Management & Stakeholders
Document Type: Project Scope & Requirements
Date: July 3, 2026

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1 Company Background

Company Name: ShopSphere

ShopSphere is an online retailer that sells Office Supplies, Furniture, and Technology products across the United States. The company operates through Consumer, Corporate, and Home Office customer segments. Although sales have grown steadily, management is concerned that profits are declining due to discounts, loss-making products, and regional performance differences.

2 Business Problem

Problem Statement

Despite increasing sales, the company's profit has been declining. Management suspects that high discounts, unprofitable products, or regional performance differences may be affecting profitability.

The goal of this project is to identify the reasons behind the declining profit and provide data-driven recommendations to improve overall business performance.

3 Business Objectives

- Increase overall profit
- Identify loss-making products
- Improve pricing strategy
- Increase customer retention
- Optimize discounts
- Improve regional performance

4 Business Questions

Management requires data-driven answers to the following key business questions:

1. Which product categories generate the highest sales?
2. Which categories generate the highest profit?
3. Which products consistently lose money?
4. Which states generate the highest profit?
5. Which states generate losses?
6. Does offering high discounts reduce profitability?
7. Who are our most valuable customers?
8. Which shipping mode performs best?
9. Which months generate the highest revenue?
10. What actions should management take?

5 Key Performance Indicators (KPIs)

The table below summarizes the core KPIs that will be tracked throughout this analysis, along with their business relevance.

KPI	Why It Matters
Total Sales	Measures revenue generated
Total Profit	Indicates overall business performance
Profit Margin	Shows profitability relative to sales
Number of Orders	Tracks business volume
Average Order Value	Measures customer spending per order
Total Customers	Indicates customer base size
Sales Growth	Tracks business growth over time
Average Discount	Measures pricing strategy effectiveness

Table 1: Key Performance Indicators tracked for the ShopSphere analysis

6 Expected Deliverables

By the end of this project, the following deliverables will be produced:

- Cleaned dataset
- SQL database
- Business SQL queries
- Exploratory data analysis (EDA)
- Tableau dashboard
- Business recommendations
- GitHub repository

Next Steps

The following phase of this project will involve data cleaning, database design, exploratory analysis, and dashboard development to answer the business questions outlined above and deliver actionable recommendations to management.